

Ryan Potechin

+1 (973) 986-8446

Boston, MA

potechin.ry@gmail.com

Work Experience

Johnson & Johnson, Raynham, MA

Manufacturing Engineer / Project Engineer / Validation Engineer

May 2025 – Present

- Serve as equipment SME and lead technical engineer for \$3M+ in capital equipment purchases within a \$100M capacity expansion project, supporting over 1.5M units per year and enabling a 300% increase in output capacity
- Own all technical aspects of custom robotic grinding equipment from end-to-end, including process characterization, DFM, ABB RobotStudio programming, equipment commissioning (URS, FAT, SAT, OEM), validation (SBD, CSV-e/IQ/OQ/PQ), reliability planning (EHS compliance, PM scheduling, calibration, OEE), and on-time machine ramp up and deployment
- Design and program robot programs in RobotStudio for 185 unique orthopedic products, achieving fully automated grinding that consistently completes material removal and finishing processes with 3mm tolerance across 1M+ parts per year
- Develop custom fixtures in Siemens NX, author engineering drawings, and partner with machine shops to fabricate designs applying DFM/DFX, material selection, and GD&T to support machine validation and optimize manufacturing processes
- Led implementation of Allen-Bradley PLC/HMI and ABB Robot Systems into a medical device manufacturing environment, driving alignment across quality, process, operations, and maintenance teams while authoring and maintaining manufacturing documentation including BOMS, DHRs, and process records in compliance with GxP/GDP, SDLC, and PLM requirements

Resonetics LLC, Westborough, MA

Development Engineering Co-op

January 2023 – July 2023

- Designed and developed catheters for applications including TMVR, imaging, and treatment of normal pressure hydrocephalus
- Performed device tensile testing, tortuous anatomy testing, and verification and validation studies to ensure clinical performance
- Collaborated with physicians and participated in animal studies to incorporate clinical feedback into device design
- Utilized SolidWorks for 3D modeling and prototyping, designing fixtures that improved process efficiency by up to 20%
- Authored SOPs for catheter manufacturing and successfully implemented them in an ISO-certified cleanroom environment

Insulet Co, Acton, MA

Systems Engineering Co-op

January 2022 – August 2022

- Developed and refined product requirement based on user needs as part of the design control process for the Omnipod 5 system
- Applied ISO 14971 risk management standards to the Omnipod 5 system and authored SHA and DFMEA updates and changes
- Led cross-functional technical reviews with quality and engineering teams to drive alignment on requirement changes

Projects

Adaptive Bocce Project, Engineering Lead

August 2023 – April 2024

- Managed a team of engineers to design and develop a Bocce ramp for the Boston Self Help Center, supporting their adaptive Bocce program and enabling people with upper arm mobility to play and compete
- Utilize SolidWorks for modeling and simulation and utilized power tools and laser cutting to fabricate working prototypes

MedCheck App Capstone Project, Project Lead

July 2023 – April 2024

- Led a team of engineers in creating a smartphone based medical checkup app by defining product requirements, researching and developing solutions, and executing a project plan, while maintaining documentation for all design decisions and changes
- Utilized MATLAB to extract and analyze PPG data from smartphone videos to track heart rate, breathing rate, and blood oxygen

Education

Northeastern University, Boston, MA

September 2024 – May 2025

Master of Science in Product Development | GPA: 3.92

Relevant Coursework: Medical Physiology, Engineering Product Design, Iterative Product Prototyping, Product Development for Engineers, Customer Driven Technical Innovation

Northeastern University, Boston, MA

September 2020 – May 2024

Bachelor of Science in Bioengineering, Concentration in Biomedical Devices and Bioimaging

Summa Cum Laude, Dean's List (2020 – 2024), Dean's Scholarship Recipient (2020 – 2024) | GPA: 3.90

Relevant Coursework: Design of Implants, Biomechanics, Biomedical Devices & Imaging, Bioengineering Measurement Experimentation and Statistics

Skills

Design & Computer: Siemens NX, SolidWorks, MATLAB, C++, AutoCAD, Arduino, RobotStudio, JIRA, Arena, Windchill

Manufacturing: 3D Printing (FDM and SLA), Laser Cutting, Mill, Lathe, Dremel, Laser Welding, Soldering, Handheld power tools

Documentation and Compliance: Technical Writing, Process Development, Validation, Regulatory compliance, IQ/OQ/PQ

Involvements

Jimmy Fund Fall Fest, Founder and Advisor

August 2022 – Present

- Organized and spearheaded the first and second Kappa Sigma Jimmy Fund Fall Fest to support the Dana-Farber Cancer Institute
- Established and managed committees for critical tasks, including sponsorships, event logistics, and event day operations
- Support and advise the organizers of the event, which has continued to grow and has raised over \$125,000 since its inception

Interests

Yoga, Ice Hockey, Cooking, Legos, Violin, Skiing, Hiking, Board Games, 3D printing, and Running/Endurance Sports